

Categorizing Fault-Tolerant Decentralized Learning along the Axes of Privacy, Robustness, and Efficiency

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Abstract: Decentralized learning allows for collaborative training of machine learning models without central coordination. By identifying building blocks for the placement of decentralized learning systems within the privacy-robustness-efficiency trilemma, we provide an overview of the state of the art, help to guide design decisions in fault-tolerant decentralized learning systems, and identify research gaps. This talk introduces our ongoing work on this classification scheme and showcases how we can use it to understand how fault-tolerant state-of-the-art decentralized learning is.

Keywords: Decentralized Learning, Security of Machine Learning

In the field of machine learning, training and inference of models has become massively centralized. Decentralized learning [Li18] aims to break this trend by distributing training among a set of nodes without central coordination. As decentralized learning is still a young and emerging field, it lacks a coherent way of categorizing new approaches.

As all kind of distributed learning adhere to the privacy, robustness, and efficiency trilemma [AGS23], this talk argues that a categorization along these axes allows us to order the field of decentralized learning. The trilemma states that no system can achieve perfect privacy, robustness, and efficiency simultaneously. In this case, privacy refers to the confidentiality of a nodes local training data. Robustness means the system’s ability to withstand the presence of Byzantine actors trying to disturb the training process. Efficiency measures the performance of the system, regarding the model utility, computational and communication complexity.

After surveying 30 current works in the field of decentralized learning, we identified a small set of building blocks used by state of the art works to achieve privacy, robustness or efficiency. For privacy, these building blocks are either secure multi-party computation based (e. g. [Zh22]) or based upon differential privacy (e. g. [XZW21]). Robustness is provided by so-called “secure-aggregation” schemes, permissioned networks (e. g. [Xu22]) or verification (e. g. [Wa25]). Building blocks for efficiency are broader, as they either optimize communication or local training.

Positioning these building blocks inside the trilemma, as in Fig. 1, lets us classify approaches to decentralized learning and assess the state of the art. The fault tolerance of decentralized learning therefore is mainly driven by secure aggregation schemes or permissioned networks. Both techniques do not provide general robustness guarantees: Secure aggregation mainly identifies outliers to the training process to hinder model poisoning, but model backdoors

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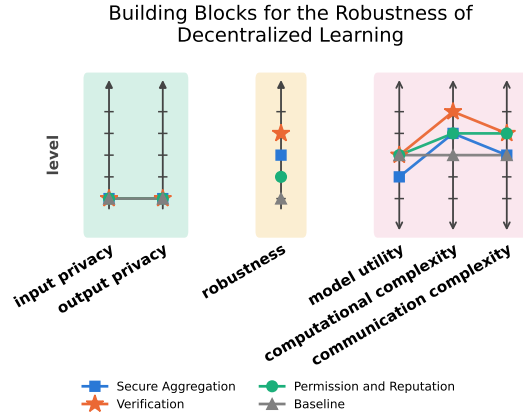


Fig. 1: Exemplary relative categorization of building blocks for robustness inside the privacy, robustness, efficiency trilemma for decentralized learning.

might still be implemented. Permissioned networks and similar techniques such as reputation scoring, transfer the ensuring of fault tolerance to other systems that might be circumvented. Verification is sound in principle, but lacks practical usability due to large computational resources needed and limited flexibility. This talk therefore argues, that further research regarding the fault tolerance of decentralized learning has to be conducted, to allow for real world use of such systems.

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